

Question Paper

Exam Date & Time: 29-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Computer Vision [IT_4408]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the image types. Explain how these image types differ in terms of the amount of information they contain and their typical applications in real-world scenarios. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Analyze the roles of pixels, color channels, feature extraction, and pattern recognition in Computer Vision using a suitable real-world scenario of your choice. Illustrate your explanation with diagrams. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Discuss how the Histogram of Oriented Gradients (HOG) descriptor represents object shape and structure. Describe the major computation steps and discuss why it performs well in pedestrian detection tasks.. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Apply the histogram equalization and also find the negative to the 4*4 matrix of your choice. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Apply calibration techniques to align multiple cameras on a robotic arm for precise 3D object localization. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze the knowledge of geometric transformations to stabilize video frames recorded from a moving vehicle by compensating for camera motion. (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyse the Lucas-Kanade method to estimate the motion of a surgical instrument in consecutive endoscopic frames. Describe how you would select window size, pyramidal levels, and error thresholds to maintain accurate tracking despite specular highlights (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Apply an optical-flow based motion compensation before background subtraction in a traffic camera to improve vehicle detection during camera vibrations. Explain parameter choices and evaluation criteria (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Analyze a face detection system for a classroom attendance application that automatically marks students as present based on camera input. Discuss challenges like lighting, occlusion, and pose variation. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply a Convolutional Neural Network (CNN) to detect cracks and corrosion in metal structures from drone inspection images. Describe how data augmentation helps improve generalization (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4408:Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4408:Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)